

Uroš Pantelić

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Software engineer focused on backend systems and ML infrastructure: tensor-parallel LLM inference at Tenstorrent (upstream to TT-XLA), model benchmarking at Microsoft, multi-tenant access control at Nextesy.

EXPERIENCE

Nextesy

May 2026 – Aug 2026

Software Engineer Intern, Backend

- Designed and implemented **role-based access control** for a multi-tenant backend, with fine-grained permissions across user roles, resources and organizational scopes.
- Reduced authorization overhead on high-traffic endpoints by resolving permissions with **indexed database queries and query batching**.
- Built an **integration and load-testing suite** that verifies access rules and benchmarks endpoint throughput under concurrent load, catching permission regressions before production.

Tenstorrent

Feb 2025 – Nov 2025

Machine Learning Intern & Open Source Contributor (TT-XLA)

- Implemented **tensor-parallel inference for LLaMA 3.1 8B** in JAX/Flax on 1×4 and 2×4 device meshes, compiled through StableHLO into Tenstorrent's **MLIR-based compiler stack (TT-XLA)**; matched PyTorch/Hugging Face outputs token-for-token and contributed the implementation upstream.
- Enabled **parameter-efficient fine-tuning (LoRA/DoRA)** for LLaMA 3.2 1B by extending training pipelines for multi-device adaptation.
- Built reproducible training and evaluation pipelines; benchmarked latency, throughput, tokens/sec and memory use across device configurations.

Microsoft

Aug 2024 – Nov 2024

Machine Learning Engineer Intern

- Prototyped **selective state-space models (Mamba)** in PyTorch and built a benchmarking harness against LLaMA/GPT-style transformer baselines.
- Measured latency, throughput, memory and quality; my technical report informed model-selection decisions.

PROJECTS

AI Agent Orchestrator (2026)

Python, tmux, Git, Claude Code

- Python CLI and daemon that runs several LLM coding agents in parallel, each in its own git worktree and tmux session, pulling tasks from a shared queue with per-task locking.
- Park/resume lifecycle: parking commits work in progress and releases the lock so any idle worker can resume it; deterministic outage recovery, self-healing of leaked locks, web UI board.

Distributed Inference Platform

Python, FastAPI, Redis, Docker, WebSockets

- Job orchestrator for CNN inference with Kubernetes-style queues and pluggable executors: dynamic worker scaling, DAG dependency resolution, prioritization and automatic retries.
- Real-time WebSocket dashboard for per-node load, throughput, queue depth and retry rate; near-linear throughput scaling under increasing concurrency.

COMPETITIONS

Qualcomm LPCVC 2025, Track 1 – 3rd Place. On-device image classification under real-world distribution shift, compiled and profiled on Snapdragon 8 Elite via Qualcomm AI Hub; **presented at CVPR 2025** (Nashville).

DigiHack – 2nd Place (predictive maintenance) • **ASEF Hackathon – 2nd Place** (air-quality forecasting)

EDUCATION

University of Belgrade, School of Electrical Engineering

2021 – 2026

B.Sc. Electrical Engineering, Signals and Systems (Computer Science electives)

GPA 9.0 / 10.0

Thesis: White-box vs. black-box targeted adversarial attacks on a CIFAR-10 ResNet-50 (FGSM, PGD, C&W, Square Attack); PGD reached 99.9% targeted success in about 3 gradient steps.

SKILLS

Languages: Python, C++, Java, TypeScript, SQL

Backend: FastAPI, Django, PostgreSQL, Redis, Docker, REST, WebSockets, CI/CD, Git

ML systems: PyTorch, JAX/Flax, Hugging Face, tensor parallelism, LoRA/DoRA, XLA/MLIR

Certifications: Deep Learning & Machine Learning Specializations (DeepLearning.AI), Google ML Bootcamp

Spoken: Serbian (native), English (C1), Turkish (basic)